Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |

AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | McLean, VA (open to SF / NYC)

PROFESSIONAL SUMMARY

AI agent engineer who **builds production multi-agent systems and evaluation frameworks** - the core of Scale AI's Frontier Agents team. Built AgentProbe end-to-end: a **3-agent LangGraph platform** (Orchestrator, RAG Tracer, Auto Fix) with tool calls, multi-step reasoning, and inter-agent handoffs - the same architecture Frontier Agents deploys for enterprise customers. **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD gates - agents are auditable in production, aligned with Scale's AI quality mission. **Multi-provider LLM routing** (Anthropic, OpenAI, Groq, Ollama), **ChromaDB RAG pipelines**, and a **15-endpoint FastAPI backend**. **8+ years Python production** at Entrata (100K+ MAU, 1.6M records, 120+ Agile sprints, 100% PHI compliance). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Multi-Agent Systems (LangGraph, Tool Calls) | LLM Evaluation & Quality Frameworks | Python (8+ years production)

RAG Pipelines & Vector Search (ChromaDB) | Prompt Engineering & LLM Routing | FastAPI / REST Platform Layer

Enterprise-Scale Delivery (100K+ MAU) | AWS / Docker / CI/CD | PHI / HIPAA Compliance Engineering

React / TypeScript / GraphQL | Cross-Functional Technical Communication | End-to-End Project Ownership

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Served as **primary technical point of contact** for cross-functional product delivery across **20+ full-stack modules** (Python, TypeScript, React, PostgreSQL, GraphQL) - partnered directly with product, engineering, and business stakeholders across **120+ Agile sprints** to translate requirements into production systems serving **100K+ monthly active users**.
- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - delivered production AI on regulated enterprise data with **100% PHI compliance, zero security incidents** using Libsodium-encrypted AWS S3 pipelines.
- Led Docker-based containerization for 30%+ of microservices, cutting execution time by **50%**; improved production query performance by **30%**; maintained **99.5% uptime** across enterprise SaaS; built 5+ debugging platform tools adopted across squads, reducing resolution time by **40%**.
- Eliminated 80% of manual operational workload by designing and automating decision-support workflows in Python and PL/SQL - consistent bug count under 10 for 6+ consecutive months.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Translated complex technical concepts for 30+ students and designed supplementary Python exercises - the same communication pattern required when onboarding enterprise customers to AI agent infrastructure.

PROJECTS

AgentProbe: Production AI Agent Platform - The Frontier Agents Architecture in Open Source OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · TypeScript · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built the same system Scale AI's Frontier Agents team deploys for enterprise customers - production AI agents from architecture to evaluation. **Agent architecture**: 3-agent LangGraph system (Orchestrator routes tasks, RAG Tracer validates retrieval quality, Auto Fix closes hallucination loops) with tool calls, failure handling, and inter-agent state handoffs. **Evaluation framework**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD gates - agents are auditable and reliable in production, aligned with Scale's AI quality mission. **LLM stack**: multi-provider routing across Anthropic, OpenAI, Groq, Ollama with dynamic selection by task and capability; ChromaDB vector store + embeddings + retrieval quality scoring. **Platform**: 15-endpoint FastAPI backend + React observability dashboard. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Healthcare Enterprise AI - HIPAA-Compliant LLMs + FHIR APIs

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · GraphQL

End-to-end enterprise AI integration for a regulated domain - NLP routing natural language queries to FHIR APIs via LLMs, BERT, and embeddings. **HIPAA-compliant RBAC. ~60% retrieval time reduction.** Demonstrates full-stack compliance-aware AI deployment - the pattern Frontier Agents customers in healthcare require. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, Tool Calls, LLM APIs (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates, Prompt Engineering, Context Engineering

Frontend & Backend: React, FastAPI, Flask, GraphQL, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum, Claude Code, GitHub Copilot, Cursor